Abhinav Anand

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Summary

Software engineer with 4+ years building developer platforms, backend integrations, and cloud automation at Red Hat and for Cisco. Production experience with Go, Python, TypeScript/React, Kubernetes, and infrastructure as code.

Experience

Red Hat India Pvt. Ltd. Oct 2024 – Present
Software Engineer Bengaluru, India

- Built a production Go Kubernetes operator to automate Ansible Automation Portal deployment on OpenShift; composed custom resources with the Red Hat Developer Hub operator and delivered Operator Lifecycle Manager packaging.
- Hardened the operator with SSRF prevention, scoped role-based access control (RBAC), network policies, and secret validation; built Tekton/Konflux end-to-end tests against ephemeral OpenShift clusters.
- Built TypeScript/React Backstage plugins for Ansible job execution, execution-environment image builds, and self-service scaffolding; implemented catalog ingestion, progressive loading, and in-memory caching.
- Implemented OAuth2 flows across source-control providers, preserving sessions through redirects and forwarding encrypted secrets to scaffolder actions; enforced permissions across backend APIs and UI components.
- Built cross-page notifications and background polling independent of component lifecycles; maintained framework upgrades and concurrent release branches with Helm deployments.
- Shipped Python 3.13 compatibility across Ansible developer tools, CLI scaffolding commands, VS Code webviews, and portal end-to-end tests through open-source contributions.

MapleLabs Solutions Pvt. Ltd. Aug 2022 – Sep 2024
Software Engineer (Contractor for Cisco) Bengaluru, India

- Automated provisioning, decommissioning, and scaling of a 40-node Cisco Tetration cluster on Google Cloud Platform using Python and Terraform across Compute Engine, IAM, storage, and VPC firewalls.
- Migrated VM-based microservices to Kubernetes and CentOS 7 systems to AlmaLinux 8; developed Python automation for cluster scaling and Ansible roles for configuration management.
- Automated a cross-data-center HDFS migration using Python for a three-DataNode cluster with approximately 3 TB stored per node and replication factor three; replaced and decommissioned one node at a time to maintain application availability.
- Built Jenkins CI/CD pipelines for builds, tests, deployments, upgrades, and infrastructure provisioning; standardized machine images and added pytest suites and pre-deployment environment validation.

Projects

Adaptive Query Accelerator | C++20, CMake | github.com/abhikdps/adaptive-query-accelerator

- Built a persistent key-value store with memory-mapped I/O, slotted pages, a hash index, and reader-writer locking; added tiered record/page caches, pluggable eviction policies, and Markov-based prefetching.
- Benchmarked 1.62M writes/sec (asynchronous disk flush), 10.4M scan records/sec, and 13.5x warm-read speedup (0.11 vs. 1.48 microseconds) on an Apple M3 with 1M records; documented reproduction steps and added recovery, concurrency, and storage tests.

Distributed Task Queue | Python, Kafka, PostgreSQL, Redis | GitHub

- Built gRPC/REST APIs and workers for task submission, status queries, exponential-backoff retries, and cancellation; added Redis task caching, Kubernetes deployment, and OpenTelemetry/Prometheus observability. Design targets: 10M+ tasks/day and sub-ms P99 latency.

Zero-Overhead Load Balancer | C, eBPF/XDP, libbpf | GitHub

- Built a Layer 4 TCP/UDP load balancer with eBPF/XDP packet forwarding, connection affinity, weighted backend selection, and incremental IP/transport checksum updates; reported approximately 300K PPS in a Lima VM (generic/SKB mode), with near-zero CPU overhead as a design goal.
- Implemented TCP health checks, backend configuration reloads without detaching the XDP program, BPF ring-buffer event streaming, and per-CPU statistics; validated packet behavior with Scapy-based functional tests.

Technical Skills

Languages: Python, Go, TypeScript, JavaScript, C++, C, Bash
Backend / Web: Microservices, REST APIs, gRPC, Node.js, Django, Tornado, React, Backstage, OAuth2, RBAC
Cloud: Linux, CI/CD, GCP, Kubernetes, OpenShift, Terraform, Docker, Helm, Ansible, Jenkins, Tekton, Konflux, GitHub Actions
Data / Telemetry: PostgreSQL, Redis, Kafka, MongoDB, OpenTelemetry, Prometheus, Grafana, Loki
Testing: pytest, Jest, Playwright, GoogleTest

Education

Acharya Institute of Technology Aug 2018 – Jul 2022
Bachelor of Engineering, Information Science CGPA: 8.4/10